

CHANDRIKA NAGA KUMARI PITTU

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LinkedIn

Summary

Embedded Systems Engineer specializing in real-time embedded software development, UAV autopilot systems, and sensor fusion algorithms. Proficient in STM32 (ARM Cortex-M) microcontrollers, Embedded C, and FreeRTOS-based system design. Experienced in low-level driver development, hardware-software integration, and real-time data processing using communication protocols such as I2C, SPI, and UART. Strong focus on building robust, scalable, and high-performance embedded solutions for aerospace and robotics applications.

Education

Bachelor of Technology (B.Tech), Electronics and Communication Engineer

2021 – 2025

Rajiv Gandhi University of Knowledge Technologies, Ongole

CGPA: 8.9

Pre-University Course (PUC)

2019 – 2021

Rajiv Gandhi University of Knowledge Technologies

CGPA: 9.98

Secondary School Certificate (SSC)

2019

Z.P.H.S High School

CGPA: 10

Experience

Embedded Engineer – Vu Dynamics (IIT Kanpur Startup)

Dec 2025 – Present

- Designed and developed embedded software for UAV systems using STM32 microcontrollers and ARM Cortex-M architecture.
- Implemented low-level drivers and peripheral interfaces for IMU, GPS, magnetometer, and communication modules like I2C, UART.
- Developed Battery Management System (BMS) using STM32 for real-time monitoring of voltage, current, and temperature parameters.
- Implemented State of Charge (SoC) estimation, fault protection mechanisms, and cell balancing algorithms for battery safety and efficiency.
- Designed real-time control systems using FreeRTOS, including task scheduling, synchronization, and inter-process communication.
- Contributed to UAV autopilot system development, implementing sensor fusion algorithms (EKF), AHRS computations, and flight stabilization logic using PID control.
- Performed extensive debugging, hardware validation, and real-time optimization to enhance system accuracy and performance.
- Integrated embedded firmware with UAV hardware systems for real-time control, data acquisition, and system-level reliability.

Autopilot Trainee – UAV Systems

Jun 2025 – Nov 2025

- Developed real-time autopilot functionalities using STM32 microcontrollers and STM32CubeIDE.
- Integrated IMU, GPS, and magnetometer sensors for attitude estimation and navigation.
- Implemented Extended Kalman Filter (EKF) algorithms for accurate state estimation.
- Designed multitasking embedded systems using FreeRTOS for control loops, sensor processing, and communication modules.
- Conducted system calibration, flight data validation, and stability testing.

Projects

UAV Autopilot System (STM32, FreeRTOS, EKF)

- Designed and implemented an autonomous UAV autopilot system using STM32 microcontrollers and Embedded C.
- Integrated IMU, GPS, and magnetometer sensors for real-time flight data acquisition and processing.
- Implemented AHRS and EKF-based sensor fusion algorithms for accurate orientation and position estimation.
- Utilized FreeRTOS for real-time task management, including sensor handling, control loops, and communication.

Wearable Posture Monitoring System

- Designed an embedded wearable system using accelerometer and gyroscope sensors to monitor user posture.
- Implemented real-time alert mechanisms using vibration and LED indicators.
- Performed sensor calibration and data processing for improved accuracy.

Technical Skills

- Programming:** C, Embedded C
- Microcontrollers:** STM32 (F4, H7), ARM Cortex-M, ATmega328P
- Tools:** STM32CubeIDE, Arduino IDE, Oscilloscope
- RTOS:** FreeRTOS (tasks, queues, semaphores)
- Protocols:** I2C, SPI, UART
- Sensors:** IMU, GPS, Magnetometer, Barometer